Additive and Abelian Categories

Lecture 1

Contents

1 Categories

Definition 1.1. A **category** $\mathcal{C}$: objects; a set $\mathrm{Mor}_{\mathcal{C}}(A, B)$ for each pair of objects A, B of $\mathcal{C}$; associative composition $\mathrm{Mor}_{\mathcal{C}}(B, C) \times \mathrm{Mor}_{\mathcal{C}}(A, B) \to \mathrm{Mor}_{\mathcal{C}}(A, C)$; identities $\mathrm{id}_A \in \mathrm{Mor}_{\mathcal{C}}(A, A)$.

Example 1.2.

- **Set**, **Grp**, **Ab**, R-**Mod**, **Top**;
- a group G, viewed as a category $\mathrm{B}G$ with one object $*$ and $\mathrm{Mor}_{\mathrm{B}G}(*, *) = G$; every morphism is invertible, so $G = \mathrm{Aut}_{\mathrm{B}G}(*) = \mathrm{End}_{\mathrm{B}G}(*)$;
- a poset: $p \leq q$ gives one arrow $p \to q$.

The last two show that morphisms need not be functions, so all definitions today must be purely arrow-theoretic.

Definition 1.3. Let $\mathcal{C}$ be a category and $f \in \mathrm{Mor}_{\mathcal{C}}(A, B)$. Then f is **mono** if for all $g, h \in \mathrm{Mor}_{\mathcal{C}}(X, A)$, $fg = fh \Rightarrow g = h$; **epi** if for all $g, h \in \mathrm{Mor}_{\mathcal{C}}(B, X)$, $gf = hf \Rightarrow g = h$; **iso** if

$$\exists\, g \in \mathrm{Mor}_{\mathcal{C}}(B, A) \quad \text{with} \quad gf = \mathrm{id}_A \quad \text{and} \quad fg = \mathrm{id}_B\,.$$

Warning. In **Ring**, the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is mono *and* epi, but not iso. Remember this pathology.

Universal properties. Terminal T: unique $X \to T$ for every X; initial is dual. Product (= universal for maps *in*) and coproduct (= universal for maps *out*):

$$\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[dr, "g"] \arrow[d, dashed, "\exists!\,\langle f,g\rangle"] & \\
A & A \times B \arrow[l, "p_A"] \arrow[r, "p_B"'] & B
\end{tikzcd}
\qquad
\begin{tikzcd}
 & X & \\
A \arrow[ur, "f"] \arrow[r, "i_A"'] & A + B \arrow[u, dashed, "\exists!\,[f,g]"] & B \arrow[ul, "g"'] \arrow[l, "i_B"]
\end{tikzcd}$$

Anything universal is unique up to unique isomorphism.

category	product	coproduct
Set	$A \times B$	$A + B$ (disjoint union)
Grp	$A \times B$	$A * B$ (free product)
Ab, R-**Mod**	$A \oplus B$	$A \oplus B$ $\leftarrow$ **same object!**

The coincidence in the last row is the seed of the whole lecture.

Duality. Reversing all arrows in a theorem yields a theorem (proved in $\mathcal{C}^{\mathrm{op}}$). Used constantly.

2 Zero objects and zero morphisms

Definition 2.1. 0 is a **zero object** if it is both initial and terminal. (**Set**: no. **Grp**, R-**Mod**, pointed sets: yes.)

Definition 2.2 (and Lemma)**.** In a category $\mathcal{C}$ with zero object, the **zero morphism** $0_{A,B} \in \mathrm{Mor}_{\mathcal{C}}(A, B)$, and its absorbing property:

$$\begin{tikzcd}
A \arrow[rr, "0_{A,B}"] \arrow[dr] & & B \\
 & 0 \arrow[ur] &
\end{tikzcd}
\qquad
\begin{tikzcd}
A \arrow[r] \arrow[rrr, bend right, "0_{A,C}"'] & 0 \arrow[r] \arrow[rr, bend left, "\text{unique map } 0 \to C"] & B \arrow[r, "f"'] & C
\end{tikzcd}$$

so $f \circ 0 = 0$ and $0 \circ f = 0$: any composite through 0 factors through the unique map $0 \to C$.

Each morphism set now has a basepoint. Next: a full addition, for free.

3 Semiadditive categories

3.1 The comparison map and biproducts

Given a zero object, products and coproducts, there is a *canonical* comparison map r, defined on the injections:

$$A + B \xrightarrow{\quad r \quad} A \times B \qquad r \circ i_A = \langle \mathrm{id}_A, 0 \rangle, \quad r \circ i_B = \langle 0, \mathrm{id}_B \rangle, \qquad r = \begin{pmatrix} \mathrm{id}_A & 0 \\ 0 & \mathrm{id}_B \end{pmatrix}.$$

category	$r : A + B \to A \times B$	iso?
Ab	$A \oplus B = A \oplus B$	✓
$\mathbf{Set}_*$	wedge $A \vee B \hookrightarrow A \times B$ (“the axes”)	×
Grp	$A * B \twoheadrightarrow A \times B$ (kill cross-commutators)	×

Definition 3.1. A **biproduct** of A and B: one object, four structure maps, product and coproduct at once —

$$\begin{tikzcd} A \arrow[r, shift left, "i_A"] & A \oplus B \arrow[l, shift left, "p_A"] \arrow[r, shift left, "p_B"] & B \arrow[l, shift left, "i_B"] \end{tikzcd} \qquad \begin{array}{l} p_A i_A = \mathrm{id}_A, \quad p_B i_A = 0, \\ p_B i_B = \mathrm{id}_B, \quad p_A i_B = 0, \end{array}$$

such that (p_A, p_B) exhibit a product and (i_A, i_B) a coproduct.

Definition 3.2. $\mathcal{C}$ is **semiadditive** $\iff$ zero object + all binary biproducts $\iff$ finite products and coproducts exist and every comparison r is an isomorphism.

3.2 The miracle: morphism sets become commutative monoids

Lemma 3.3 (Eckmann–Hilton)**.** Let M be a set with two binary operations $*$ and $\star$ (no associativity assumed) and elements $e, e' \in M$ such that

1. The element e is a two-sided unit for $*$, and e' is a two-sided unit for $\star$:

$$e * a = a = a * e, \qquad e' \star a = a = a \star e' \qquad (a \in M);$$

2. The **interchange law** holds:

$$(a \star b) * (c \star d) \;=\; (a * c) \star (b * d) \qquad (a, b, c, d \in M).$$

Then $e = e'$, the two operations coincide, and the resulting single operation is commutative and associative. In particular $(M, *)$ is a commutative monoid.

Proof. Only the unit laws and interchange are used; each step below applies exactly one of them.

Step 1: the units coincide.

$$e \;=\; e * e \;=\; (e \star e') * (e' \star e) \;\overset{(2)}{=}\; (e * e') \star (e' * e) \;=\; e' \star e' \;=\; e'.$$

(The second and fourth equalities use the unit laws for $\star$ and $*$ respectively.) Write e for the common unit.

Step 2: the operations coincide and are commutative. For all $a, b \in M$:

$$a * b \;=\; (a \star e) * (e \star b) \;\overset{(2)}{=}\; (a * e) \star (e * b) \;=\; a \star b,$$

$$a * b \;=\; (e \star a) * (b \star e) \;\overset{(2)}{=}\; (e * b) \star (a * e) \;=\; b \star a.$$

Together: $a * b = a \star b = b \star a = b * a$.

Step 3: associativity. Writing $*$ for the (now unique) operation:

$$a * (b * c) \;=\; (a \star e) * (b \star c) \;\overset{(2)}{=}\; (a * b) \star (e * c) \;=\; (a * b) * c. \qquad \square$$

Remark 3.4. The lemma is stronger than it looks: neither operation was assumed associative or commutative, yet both properties are forced. Classic corollaries elsewhere in mathematics: a monoid object in the category of monoids is a commutative monoid; the higher homotopy groups π_n $(n \geq 2)$ are abelian; the fundamental group of a topological group is abelian.

Theorem 3.5. Let $\mathcal{C}$ be a semiadditive category. Then each $\mathrm{Mor}_{\mathcal{C}}(A, B)$ is canonically a commutative monoid with unit $0_{A,B}$, and composition is bilinear.

Construction. For $f, g : A \to B$ define $f + g$ as the composite

$$A \xrightarrow{\quad\Delta\quad} A \oplus A \xrightarrow{\quad f\oplus g\quad} B \oplus B \xrightarrow{\quad\nabla\quad} B$$

where $\Delta = \langle \mathrm{id}, \mathrm{id} \rangle$ (diagonal, via the product structure) and $\nabla = [\mathrm{id}, \mathrm{id}]$ (codiagonal, via the coproduct structure).

Why commutative, associative, unital? By Lemma 3.3. The product structure alone defines one operation $+$; the coproduct structure another, $\boxplus$. Both are unital with unit 0, and they satisfy the interchange law

$$(f + g) \boxplus (h + k) \;=\; (f \boxplus h) + (g \boxplus k).$$

Lemma 3.3 then forces $+ = \boxplus$, commutative and associative. Bilinearity of composition follows from naturality of Δ and ∇:

$$\begin{tikzcd} A \arrow[r, "\Delta_A"] \arrow[d, "h"'] & A \oplus A \arrow[d, "h\oplus h"] \\ B \arrow[r, "\Delta_B"'] & B \oplus B \end{tikzcd} \qquad \begin{tikzcd} A \oplus A \arrow[r, "\nabla_A"] \arrow[d, "h\oplus h"'] & A \arrow[d, "h"] \\ B \oplus B \arrow[r, "\nabla_B"'] & B \end{tikzcd}$$ □

In R-**Mod**, unwind the construction: $a \mapsto (a, a) \mapsto (f(a), g(a)) \mapsto f(a) + g(a)$. The usual addition — *recovered, not chosen.*

Slogan. Semiadditivity is a property of a category, not a structure on it.

Theorem 3.6 (Converse)**.** If $\mathcal{C}$ has finite products and an enrichment in commutative monoids, then products are biproducts: set $i_A = \langle \mathrm{id}, 0 \rangle$, $i_B = \langle 0, \mathrm{id} \rangle$; the map induced by $f : A \to X$, $g : B \to X$ is $fp_A + gp_B$. Moreover the enrichment, if it exists, is unique.

Matrix calculus. In a semiadditive category $\mathcal{C}$, morphisms $\bigoplus_j A_j \to \bigoplus_i B_i$ are the same as matrices $(f_{ij} : A_j \to B_i)$, and composition is matrix multiplication. Linear algebra notation is not an analogy; it is the structure of the category.

Example 3.7 (Semiadditive but *not* additive)**.**

- **CMon**, commutative monoids;
- natural numbers as objects, matrices over $\mathbb{N}$ as morphisms;
- **Rel**, sets and relations: here $f + f = f$, so inverses cannot exist.

4 Additive categories

Definition 4.1. A category $\mathcal{C}$ is **additive** if it is semiadditive and every morphism monoid $\mathrm{Mor}_{\mathcal{C}}(A, B)$ is a group; equivalently, $\mathcal{C}$ is **Ab**-enriched with finite biproducts.

Example 4.2.

- **Ab**, R-**Mod**, $\mathbf{Vect}_k$;
- free modules; projective modules (a full subcategory closed under $\oplus$ and containing 0 stays additive);
- Banach spaces; chain complexes; vector bundles.

Non-examples:

- **Grp**, $\mathbf{Set}_*$ (no biproducts);
- **CMon**, **Rel** (no inverses).

Remark 4.3 (One object = ring)**.**

$$\Big\{\text{additive category } \mathcal{C} \text{ with one object } *\Big\} \longleftrightarrow \Big\{\text{ring } R = \mathrm{End}_{\mathcal{C}}(*) = \mathrm{Mor}_{\mathcal{C}}(*, *)\Big\}$$

Additive categories are "rings with several objects" (Mitchell); Morita theory, ideals, radicals all generalize.

Proposition 4.4. For a functor $F : \mathcal{A} \to \mathcal{B}$ between additive categories, the following are equivalent:

1. $F(f + g) = Ff + Fg$ on each $\mathrm{Mor}_{\mathcal{A}}(A, B)$;
2. F preserves biproducts;
3. F preserves finite products;
4. F preserves finite coproducts.

(Property, not structure — again.) Examples: $\mathrm{Mor}_{\mathcal{A}}(M, -) : \mathcal{A} \to \mathbf{Ab}$, its contravariant sibling $\mathrm{Mor}_{\mathcal{A}}(-, N)$, and $M \otimes_R - : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$.

Why keep going? In the additive category of free abelian groups, $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ has no cokernel. Additive $\neq$ ready for homological algebra.

5 Kernels and cokernels

Definition 5.1. In a category $\mathcal{C}$ with zero object, a **kernel** of $f \in \mathrm{Mor}_{\mathcal{C}}(A, B)$ is a morphism $k : K \to A$ with $fk = 0$ such that every $g : X \to A$ with $fg = 0$ factors uniquely through k (left); a **cokernel** is a morphism $c : B \to C$ with $cf = 0$ such that every $h : B \to X$ with $hf = 0$ factors uniquely through c (right):

$$\begin{tikzcd} K \arrow[r, "k"] & A \arrow[r, "f"] & B \\ X \arrow[u, dashed, "\exists!"] \arrow[ur, "g"'] & & \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, "c"] \arrow[dr, "h"'] & C \arrow[d, dashed, "\exists!"] \\ & & X \end{tikzcd}$$

Example 5.2. R-**Mod**: $\ker f = f^{-1}(0)$, $\operatorname{coker} f = B/\operatorname{im} f$. **Grp**: $\operatorname{coker} f = B/(\text{normal closure of } \operatorname{im} f)$.

Lemma 5.3. In any category $\mathcal{C}$ with zero object, kernels are monos and cokernels are epis. In an *additive* category $\mathcal{C}$, f is mono $\iff$ $\ker f = 0$, since $fg = fh \iff f(g-h) = 0$ — subtraction earns its keep here.

Definition 5.4. A mono is **normal** if it *is* a kernel of some morphism; an epi is **conormal** if it *is* a cokernel.

Invisible in R-**Mod** (every submodule is a kernel); visible elsewhere:

category	non-normal mono	symptom
Grp	$\langle(12)\rangle \hookrightarrow S_3$	kernels = *normal* subgroups only
torsion-free abelian groups	$\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ (its coker = 0)	mono + epi, not iso
Banach spaces	$\ell^1 \hookrightarrow \ell^2$ (dense, not closed)	mono + epi, not iso

The abelian axioms are engineered to outlaw exactly this column.

6 Abelian categories

Definition 6.1. $\mathcal{A}$ is **abelian** if

1. It is additive.
2. Every morphism has a kernel and a cokernel.
3. Every mono is normal and every epi is conormal.

Self-dual: $\mathcal{A}$ abelian $\iff$ $\mathcal{A}^{\mathrm{op}}$ abelian; every theorem dualizes for free. (Remarkably, additivity in (1) is automatic from the rest — even additive inverses are forced. Property, again.)

Example 6.2.

- **Ab**, R-**Mod**, $\mathbf{Vect}_k$, finite abelian groups;
- representations of groups and quivers;
- chain complexes in any abelian $\mathcal{A}$;
- sheaves of abelian groups; quasi-coherent sheaves (Grothendieck, Tôhoku 1957 — the motivating case);
- functor categories $\mathrm{Fun}(\mathcal{C}, \mathcal{A})$, pointwise.

non-example	failing axiom	witness
free / projective modules	(2)	$\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ has no cokernel
torsion-free abelian groups	(3)	$\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ non-normal mono
Banach spaces	(3)	$\ell^1 \hookrightarrow \ell^2$
filtered vector spaces	(3)	id with finer filtration on source

6.1 First consequences

Proposition 6.3 (mono + epi $\Rightarrow$ iso)**.** Let $\mathcal{A}$ be an abelian category. A morphism of $\mathcal{A}$ that is both mono and epi is an isomorphism.

Proof. f mono $\overset{(3)}{\Longrightarrow}$ $f = \ker g$ for some g. Then $gf = 0 = 0f$, and f epi gives $g = 0$. But $\ker(0 : B \to C) = \mathrm{id}_B$, so f is an iso. The $\mathbb{Z} \to \mathbb{Q}$ / $\ell^1 \to \ell^2$ pathology: abolished, by axiom (3) precisely. □

Proposition 6.4 (Image factorization = first isomorphism theorem)**.** In an abelian category $\mathcal{A}$, every $f \in \mathrm{Mor}_{\mathcal{A}}(A, B)$ factors, uniquely up to unique iso, as an epi followed by a mono:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, two heads, "e"'] & B \\ \operatorname{coim} f \arrow[r, "\cong"'] & \operatorname{im} f \arrow[u, tail, "m"'] \end{tikzcd} \qquad \begin{aligned} \operatorname{im} f &:= \ker(\operatorname{coker} f), \\ \operatorname{coim} f &:= \operatorname{coker}(\ker f). \end{aligned}$$

Sketch. $(\operatorname{coker} f) \circ f = 0$, so $f = me$ for a unique e; normality shows e is epi; the comparison $\operatorname{coim} f \to \operatorname{im} f$ is mono + epi, hence iso by the previous proposition. In R-**Mod** this reads $A/\ker f \cong \operatorname{im} f$ — now one proof for modules, sheaves, representations, complexes. □

6.2 Exact sequences

Definition 6.5. A composable pair with $gf = 0$ is **exact at** B if the induced mono $\operatorname{im} f \to \ker g$ is an iso. A **short exact sequence**:

$$0 \longrightarrow A \overset{f}{\rightarrowtail} B \overset{g}{\twoheadrightarrow} C \longrightarrow 0 \qquad f = \ker g \quad \text{and} \quad g = \operatorname{coker} f.$$

Subobjects and quotients determine each other: $A \rightarrowtail B$ extends to $0 \to A \to B \to B/A \to 0$, and dually. (This tight sub/quotient duality is what fails in **Grp**.)

Definition 6.6. An additive functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories is

- **exact** if it preserves short exact sequences;
- **left exact** if it preserves kernels;
- **right exact** if it preserves cokernels.

$\mathrm{Mor}_{\mathcal{A}}(M, -),\ \mathrm{Mor}_{\mathcal{A}}(-, N) : \mathcal{A} \to \mathbf{Ab}$:	left exact	failure measured by Ext
$M \otimes_R - : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$:	right exact	failure measured by Tor

Homological algebra = the calculus of this failure. Abelian categories = where it runs.

7 Outlook

Diagram lemmas hold: Five Lemma, Nine Lemma, the long exact sequence, and the

Snake Lemma. Given the solid commutative diagram with exact rows, there is a natural connecting morphism ∂ and a six-term exact sequence:

$$\begin{tikzcd}
& \ker a \arrow[r] \arrow[d, tail] & \ker b \arrow[r] \arrow[d, tail] & \ker c \arrow[d, tail] \arrow[dddll, dashed, rounded corners, "\partial"] & \\
& A \arrow[r] \arrow[d, "a"'] & B \arrow[r] \arrow[d, "b"] & C \arrow[r] \arrow[d, "c"] & 0 \\
0 \arrow[r] & A' \arrow[r] \arrow[d, two heads] & B' \arrow[r] \arrow[d, two heads] & C' \arrow[d, two heads] & \\
& \operatorname{coker} a \arrow[r] & \operatorname{coker} b \arrow[r] & \operatorname{coker} c &
\end{tikzcd}$$

$$\ker a \longrightarrow \ker b \longrightarrow \ker c \xrightarrow{\partial} \operatorname{coker} a \longrightarrow \operatorname{coker} b \longrightarrow \operatorname{coker} c$$

Proofs: either purely arrow-theoretically (chasing "generalized elements" $X \to A$), or via:

Theorem 7.1 (Freyd–Mitchell)**.** Every small abelian category $\mathcal{A}$ admits a fully faithful exact functor $\mathcal{A} \hookrightarrow R\text{-}\mathbf{Mod}$ for some ring R.

Consequence: finite diagram lemmas may be checked by honest element chases in modules, then transported back.

Grothendieck categories. Abelian + coproducts + exact filtered colimits (AB5) + generator $\Rightarrow$ enough injectives $\Rightarrow$ derived functors exist (sheaf cohomology!). The actual point of Tôhoku: group cohomology, sheaf cohomology, Ext — one construction.

Onward.

$$\text{abelian } \mathcal{A} \longrightarrow \mathrm{Ch}(\mathcal{A}) \longrightarrow K(\mathcal{A}) \longrightarrow D(\mathcal{A}) \longrightarrow \substack{\text{triangulated /} \\ \text{stable } \infty\text{-categories}}$$

where "stable," like "additive," is again a *property*, not structure. The theme of this lecture scales all the way up.

level	requires	gains	canonical failure below
zero object	0 initial = terminal	zero morphisms	**Set**
semiadditive	biproducts ($\times = +$)	$+$, matrix calculus	**Grp**, $\mathbf{Set}_*$
additive	inverses (**Ab**-enriched)	$-$; ker detects mono	**CMon**, **Rel**
abelian	ker, coker; (co)normality	images, 1st iso thm, exactness	tf groups, Banach

Exercises

Exercise 1. Show that kernels are monos, and that two kernels of the same morphism are canonically isomorphic.

Exercise 2. Verify the interchange law for the two operations $+$ and $\boxplus$ on $\mathrm{Mor}_{\mathcal{C}}(A, B)$, $\mathcal{C}$ semiadditive, directly from the biproduct axioms, completing the proof of Theorem 3.5. Then prove, using Lemma 3.3, that the fundamental group of a topological group is abelian.

Exercise 3. Show that in an additive category $\mathcal{C}$, a morphism $f \in \mathrm{Mor}_{\mathcal{C}}(A, B)$ is mono $\iff$ $(fg = 0 \Rightarrow g = 0)$. Where is subtraction used?

Exercise 4. In the category of torsion-free abelian groups, compute ker and coker of $\mathbb{Z} \xrightarrow{n} \mathbb{Z}$, and exhibit a morphism f whose comparison map $\operatorname{coim} f \to \operatorname{im} f$ is not an iso.

Exercise 5. Prove that $\mathrm{Fun}(\mathcal{C}, \mathcal{A})$ is abelian when $\mathcal{A}$ is, with kernels, cokernels, and exactness all computed pointwise.

Exercise 6 (Harder)**.** Prove that a semiadditive category satisfying axioms (2) and (3) of Definition 6.1 is automatically additive: additive inverses are forced.

References

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